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Test Procedure for Scintillation Radiation Detectors

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1.- OBJECTIVE AND SCOPE

1.1.- Objective

The objective of this procedure is to describe the steps to verify the electrical characteristics and the radiation response of systems with scintillation detectors employed in counting and monitoring of ionizing radiation.

1.2.- Scope

This procedure is applicable only for the verification and diagnostic of scintillation radiation detectors.

2.- NOTATION AND DEFINITIONS

2.1.- Notation

PMT	Photo-multiplier tube
MCA	Multichannel Analyzer

2.2.- Definitions

2.2.1. Detector

A device that converts the energy of a photon or incident particle in an electric pulse.

2.2.2. Scintillation Detector

A radiation detector that is made with a scintillation crystal optically coupled to a photo-multiplier tube.

2.2.3. Scintillation Crystal

A crystal that has the property of emitting light when interacts with an incident radiation, whether photons or particles. The more often used material as scintillation crystal is sodium iodine activated with thallium. NaI (Tl)

2.2.4. Photo-multiplier Tube

Electronic Vacuum device that is light sensitive, it consist of a photo-cathode, a collection system for photo electrons, several stages of multiplication by secondary emission connected to dynodes and one anode.

2.2.5 Detector Background Counting

Counting that is present even without a radiation source, when the detector is biased with a certain voltage, typically of some hundred volts.

2.2.6. Dead Time

Time Interval after the occurrence of a pulse when the detector remains insensible to the radiation.

2.2.7. Activity

Number of disintegrations per second of a radioactive source, the unit of activity is the Becquerel (Bq) that corresponds to one disintegration per second. The Curie (Ci) is also used and equals to 3.7×10^{10} Bq.

2.2.8. Exposure Rate

Amount of radiation measured as its capability to ionize the air. The utilized unit is the Roentgen/hour (R/hr).

2.2.9. Sensitivity

It is the ratio between the number of counts per unit of time obtained in a detector and the exposure rate for a given radiation source. The employed isotope must be specified.

2.2.10. Resolution

Capability of a radiation detector to distinguish between two peaks of next energies. It is expressed in % for scintillation detectors.

3.- DEVELOPMENT.

The sequence of steps to verify the electrical characteristics and radiation response of Scintillation detectors is described in the next paragraphs. A flux diagram of the process is shown in Annex I.

3.1- Generalities

A technical report of the verification of the electrical characteristics and the response to radiation of the Scintillation detectors must include the circuit diagram, environmental conditions, geometry of the testing set-up, count rate, isotope and test instruments employed.

Whenever possible, refer to the test conditions recommended by the detector manufacturer in the specifications sheet.

It is assumed that the PMT has a voltage divider as bias circuit for every dynode with the proper characteristics.

3.2.- Test Instruments

All the instruments employed in the tests must be calibrated and with a valid calibration certificate.

3.2.1 Detector Bias Power Supply.

Generally it is a high voltage power supply with enough current capability to feed the detector without any loose in regulation, the regulation must be less than 0.01 %, because the size of the pulse depends strongly on the applied voltage, the power supply must have a very good stability. The ripple should be less than 100 mV.

3.2.2 Pulse Amplifier.

A spectroscopy amplifier with shaping times less than 3 μs is required.

3.2.3 Oscilloscope

Use a digital oscilloscope with a high voltage coupling capacitor when needed.

3.2.4 Counter or Scaler

The number of counts per time unit is measured with a counter (scaler) or rate meter, for nuclear pulses, if a counting system is assembled, it generally includes a voltage discriminator to block low amplitude noise pulses. The measured pulsed in the scaler must be positive pulses.

3.2.5 Rate Meter

The average number of counts per unit of time is measured with a rate meter for nuclear pulses. The measured pulsed in the Rate Meter must be positive pulses.

3.2.6. Multichannel Analizer, MCA

A MCA with at least 1024 channels is needed to get the energy spectrum. Consider that the analog input pulse to the MCA must be positive.

3.3.- Test Conditions

3.3.1.- Background Radiation

Be sure that the only contribution to the detector counting is the natural background, avoiding the contribution due to any additional radioactive source.

3.3.2.- Temperature

Some characteristics of Scintillation detectors are temperature dependant, a reference temperature between 20º C and 25° C is recommended.

3.3.3.- Radioactive source employed

Generally, γ radiation from a Cs-137 source is specified to characterize the Scintillation detectors. A source with an activity of around 37000 Bq (1 μCi) could be used.

The response of Scintillation detectors to β radiation can be reported under specified special conditions, in this case a Sr-90 source could be used.

3.4.- Test Circuits

Whenever possible, refer to the test conditions recommended by the detector's manufacturer in the specifications sheet, if it is not available, use the diagrams shown in Fig. 1, Fig. 2 and Fig. 3.

Specify the value of the following components:

Ca : coupling capacitor

Circuit employed, kind of preamplifier or equivalent circuit (Input Resistance and capacitance).

Fig. 1 shows an example of a circuit in which the amount of pulses delivered by the detector is measured in a nuclear discriminator-counter, use the unipolar output of the amplifier.

The discriminator eliminates the electric noise, a clock in the counter fixes the counting time at the experiment.

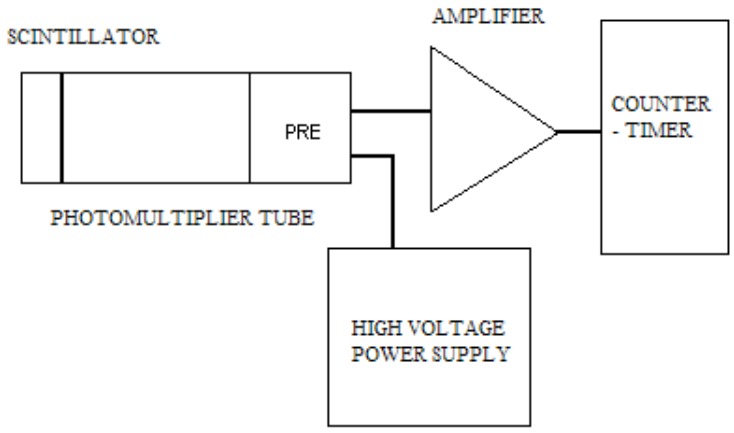


Fig.1.- Circuit employed for the measurement of Scintillation detectors with a counter timer.

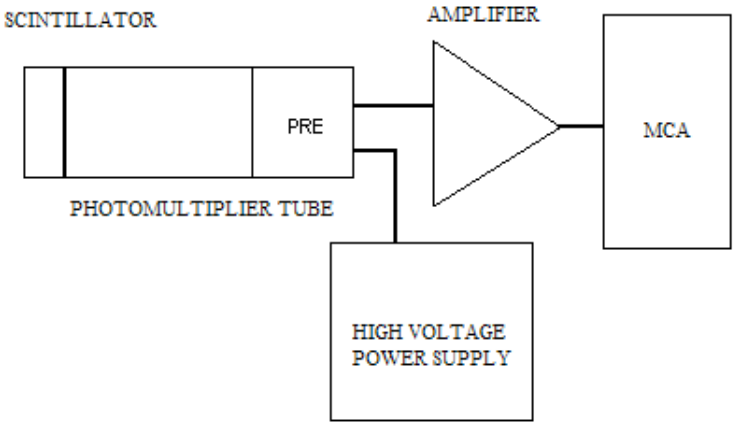


Fig. 2.- Circuit diagram for the measurement of Scintillation detectors with a MCA.

3.5.- Measurements

In this paragraph, the scintillation detector measuring parameters are defined, considering that the detector is operating in the more adequate operating point.

3.5.1 Determination of the operating voltage.

Conect the detector as shown in Fig.1. Verify the correct polarity of the high voltage for the PMT.

Put a Cs-137 radioactive source close to the detector and apply the voltage to the detector in steps of 100V until some pulses appear in the oscilloscope screen at the amplifier output, this value is V_i.
Get from the data sheet the value of the maximum applied voltage, V_M.
The opperating voltage V_o, is around the value:

$$V_0 = \frac{V_M + V_i}{2}$$

3.5.2 Measurement of the background with shielding.

Obtain the background reading in counts per minute, utilizing a circuit similar to the one shown in Fig. 1, with a voltage V_o, use a shielding on the detector (50 mm of lead plus 3 mm of aluminum). In this example, **65 c.p.m.**

3.5.3 Sensitivity Measurement.

The sensitivity to γ radiation is obtained from a Cesium-137 source, use a circuit like the shown in Fig. 1, with a bias voltage V_o. Divide the number of counts per second obtained by the activity A of the source in μCi. In this example, 900 c.p.s. /μCi
This value could be used as a reference point to determine any change in the characteristics of the detector. Keep the same detector–source geometry. In a long time comparison, consider the correction by mean life of the employed radioactive source.

3.5.4 Energy Resolution *R*, of the detector.

Use a Cesium-137 source and a set-up like the one shown in Fig. 2. Be sure that the counts be less than 5000 counts per second to avoid errors by pile-up or dead time.
Adjust the shaping time of the amplifier to 3 μs or less and adjust the pole-zero for the best unipolar signal at the output of the amplifier.
Adjust the amplifier gain or the bias voltage to have a pulse of 4 V aproximately, at the output of the amplifier for the Cs-137 peak.
Accumulate in the MCA to get more than 10 000 counts in the peak, in this condition the relative uncertainty in the results is less than 1 %.
Calibrate the MCA in energy employing the K line of the Ba with an energy of 32 keV and the main Cs-137 peak of 661.5 keV, this calibration will be not exact due to non linearities of the detector in the low energy region, but the results obtained are enough to get a first evaluation of the behaviour of the detector.
Calculate the resolution by using the definition of Fig. 3.:

$$R\ (\%) = \frac{FWHM}{H_O} \times 100$$

where:

FWHM = full width at half of maximum in keV
Ho = energy of the reference peak (Cs137 = 661.5 keV).

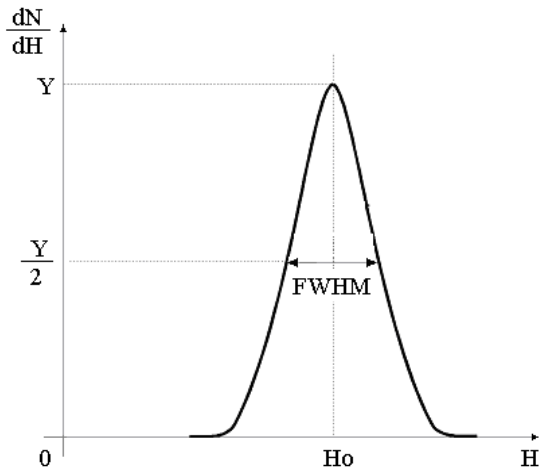


Fig. 3.- Definition of energy resolution.

In the Fig. 4, there is an example of the energy spectrum for a Cs-137 source obtained with a scintillation detector in a MCA with 1024 channels. The two peaks of the Cs-137 source are shadowed.

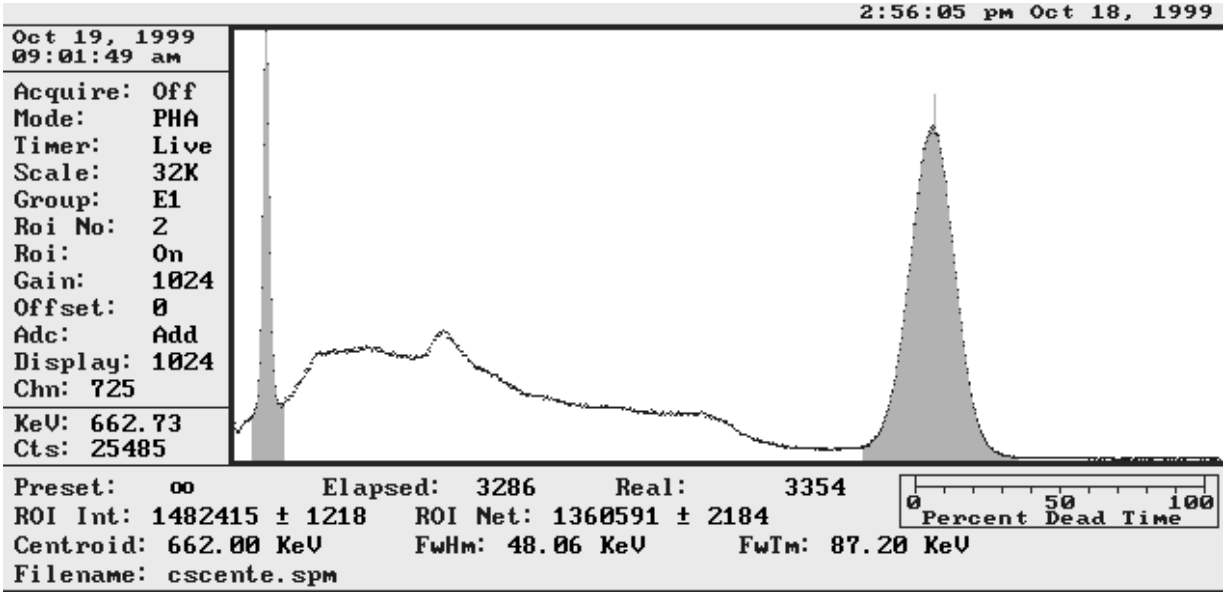


Fig. 4.- Energy spectrum of a Cs-137 source obtained with a scintillation detector.

From the Fig.4, the resolution is :

$$R\,(\%) = \frac{48.06\ keV}{661.5\ keV} \times 100$$

$$R = 7.26\ \%$$

3.6.- Pulse Characteristics

The pulse characteristics can be measured with a circuit as the one shown in Fig.1, by using an oscilloscope at the output of the preamplifier, the detector is biased at the optimum operating point.

3.6.1 Amplitude

Is the voltage measured from the base line to the maximum value of the pulse. In the example, 100 mV.

3.6.2 Rise Time

Elapsed time from 10% to 90% of the maximum value, measured in the leading edge of the pulse. In the example, 100 ns.

3.6.3 Decay time

The decay time is the elapsed time from 10% to 90% of the maximum value, measured in the trailing edge of the pulse. In the example, 50 μs.

3.7.- Detector Aging.

The aging of Scintillation detectors affects its detection characteristics and parameters. With respect to the functionality, the more important effects are:

- Drastic reduction in the pulse size due to a loss of gain in the PMT or loss of transmission of light through the scintillation crystal.
- Increase of counts respect to the expected value of sensitivity. This effect is true for detectors with a broken scintillation crystal.
- Increase of background counts. This is due to a leakage of light or aging of the PMT.

4.- ADMINISTRATION OF THE TECHNICAL REPORTS

4.1 Numbering of the reports.

All the generated technical reports must have a unique and consecutive number.

4.2.- Personnel.

The test of Scintillation detectors must be done by trained personnel.

4.3.- Technical Report of results.

4.3.1 The results of the test of Scintillation detectors must be registered in a unique technical report, stating the description of the detector, mark, model, serial number, and all the test conditions, including the name of the person who made the tests.

4.3.2 All the technical reports must be classified and keep in a folder for future consult.

5.- ACTION IN CASE OF NON CONFORMITIES.

5.1 Technical Report.

Even in the case that results of the test are not as expected, a technical report has to be elaborated, indicating the non conformities and how far are the measured characteristics from the ideal ones.

5.2 Labelling.

The components or equipments that are not under specifications or with a failure have to be marked with a label indicating: **OUT OF SPECIFICATIONS** and **FAILURE** respectively.

6.- RESPONSIBILITIES

6.1. Head of the Department.

Supervise that all the activities for testing of Scintillation detectors follow the established procedure.

6.2. Area Responsible.

6.2.1	Assure that all the electronic test equipment be in good operational conditions.
6.3.2	Verify that all the activities for testing of Scintillation detectors follow the established procedure
6.2.3	Verify that the technical reports contain all the details of the testing of Scintillation detectors.
6.2.4	Maintain a register and control of the technical reports for all the Scintillation detectors tested in the laboratory.

6.3 Operative Personnel.

6.3.1	Verify that all the electronic test equipment be in good operational conditions.
6.3.2	Follow the steps established in this procedure for the testing of Scintillation detectors.
6.3.3	Elaborate the technical report of all the tests of the scintillation detector.
6.3.4	Inform to the Area Responsible of any anomalous condition encountered during the test procedure.

7.- BIBLIOGRAPHY

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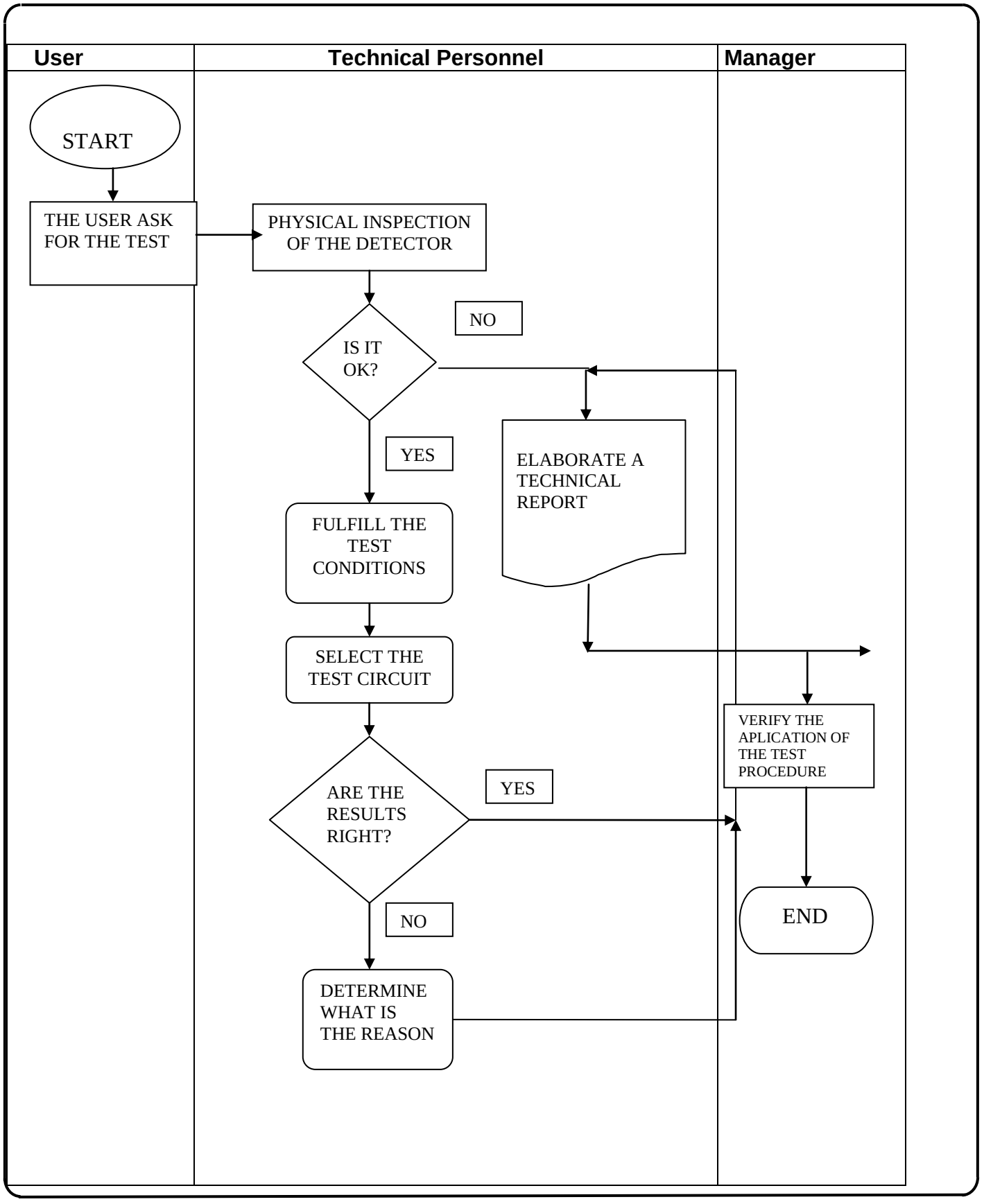
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8.- ANNEXES

Annex I

Flow Chart.


User
Technical Personnel
Manager
START
THE USER ASK FOR THE TEST
PHYSICAL INSPECTION OF THE DETECTOR
IS IT OK?
NO
YES
ELABORATE A TECHNICAL REPORT
FULFILL THE TEST CONDITIONS
SELECT THE TEST CIRCUIT
ARE THE RESULTS RIGHT?
YES
NO
DETERMINE WHAT IS THE REASON
VERIFY THE APLICATION OF THE TEST PROCEDURE
END

Annex II

Test report

TEST REPORT Nº SC-_____

Scintillation detector number:_____

Mark:_____ Model:_____
Serial Number:_____Window:_____

Physical Revision

Condition	YES	NO
Damage in the Body		
Damage in the divider chain		
Strikes		
Integrity of the window		
Corrosion		
Oxidation		

Instruments Employed

Instrument	Mark	Model	Serial number
High Voltage Power Supply			
Voltmeter			
Oscilloscope			
Counter, Scaler			
Clock, Timer			
Rate Meter			
MCA			

Environmental Conditions

Background	Temperature

Radioactive Source

Source	Energy	Activity	Date

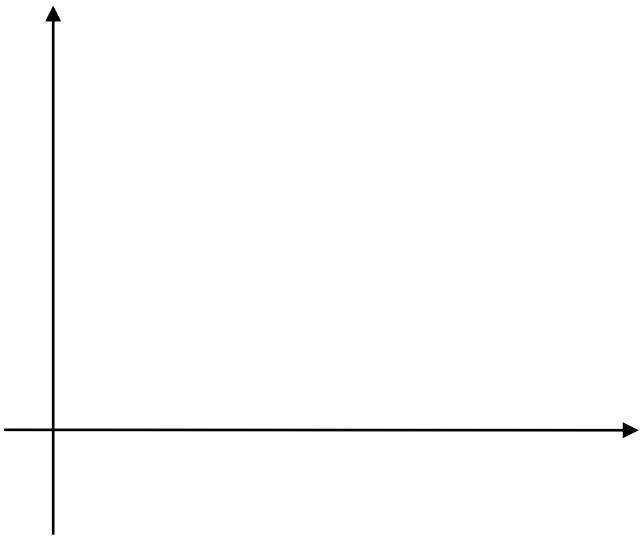
Test Circuit

Fig. 1.a	
Fig . 1.b.	

R1(MΩ)	
Cn (pF)	

Response as function of applied voltage

Bias Voltage	Count/100s



Starting voltage V_i(V)		
Maximum voltage V_M(V)		
Optimal Operating voltage V_o (V)		
Background with shielding (Counts/minute)		
Isotope:	Sensitivity (cps/μCi) / distance	_____ / ____cm
	Pulse amplitude (V)	
Rise time (ns)		
Dead time(μs)		

Diagnostic or Comments:

Tested By:	
Date:	